The documentation and process conversion measures necessary to comply with this revision shall be completed by 29 July 1994

INCH-POUND

MIL-S-19500/559B
29 April 1994
SUPERSEDING
MIL-S-19500/559A
13 September 1990

MILITARY SPECIFICATION

SEMICONDUCTOR DEVICE, UNITIZED, NPN, SILICON, SWITCHING, FOUR TRANSISTOR ARRAY
TYPES 2N6989, 2N6989U, AND 2N6990, JANTX, JANTXV, AND JANS

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the detail requirements for NPN, silicon, switching transistors in a four independent chip array. Three levels of product assurance are provided for each device type as specified in MIL-S-19500.

1.2 Physical dimensions. See 3.2 (14 pin dual-in-line, 14 pin flat package, and 20 pin surface mount).

1.3 Maximum ratings. 1/

TYPE	P_T T_A = +25°C 2/	V_{CBO} 3/	V_{EBO} 3/	V_{CEO} 3/	I_C 3/	T_{OP} and T_{STG}
	W	V dc	V dc	V dc	mA dc	°C
2N6989	1.5	75	6	50	800	-65 to +200
2N6989U	1.0	75	6	50	800	-65 to +200
2N6990	0.4	75	6	50	800	-65 to +200

1/ Maximum voltage between transistors shall be ≥ 500 V dc.
2/ Derate linearly 8.57 mW/°C above T_A = +25°C for 2N6989 and 2N6989U.
Derate linearly 2.286 mW/°C above T_A = +25°C for 2N6990. Ratings apply to total package.
3/ Ratings apply to each transistor in the array.

1.4 Primary electrical characteristics. Characteristics apply to each transistor in the array.

Limits	h_{FE2} 1/ V_{CE} = 10 V dc I_C = 1.0 mA dc	h_{FE4} 1/ V_{CE} = 10 V dc I_C = 150 mA dc	C_{obo} V_{CB} = 10 V dc I_E = 0 100 kHz ≤ f ≤ 1 MHz	Switching	
				t_{on} see fig.7	t_{off} see fig.8
			pF	ns	ns
Min	75	100			
Max	325	300	8	35	300

Limits	$\lvert h_{FE} \rvert$ V_{CE} = 20 V dc I_C = 20 mA dc f = 100 MHz	$V_{CE(sat)2}$ 1/ I_C = 500 mA dc I_B = 50 mA dc	$V_{BE(sat)2}$ 1/ I_C = 500 mA dc I_B = 50 mA dc
		V dc	V dc
Min	2.5		
Max	8.0	1.0	2.0

1/ Pulsed (see 4.5.1).

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, Defense Electronics Supply Center, ATTN: DESC-ELDT, 1507 Wilmington Pike, Dayton, OH 45444-5765, by using the Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A

FSC 5961

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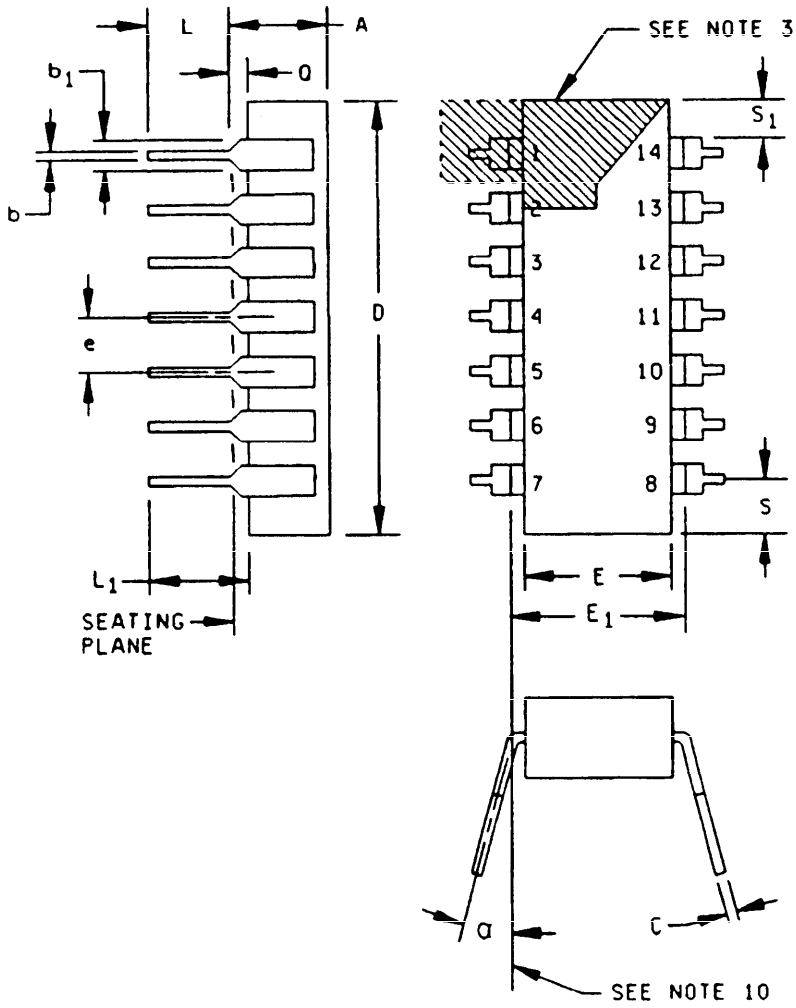


FIGURE 1. Dimensions and configuration for type 2N6989.

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Symbol	Dimensions				Notes
	Inches		Millimeters		
	Min	Max	Min	Max	
A		.200		5.08	
b	.014	.023	.36	.58	10
b_1	.030	.070	.76	1.78	4,10
c	.008	.015	.20	.38	10
D		.785		19.94	6
E	.220	.310	5.59	7.87	6
E_1	.290	.320	7.37	8.13	9

Symbol	Dimensions				Notes
	Inches		Millimeters		
	Min	Max	Min	Max	
e	.100 BSC		2.54 BSC		7,11
L	.125	.200	3.18	5.08	
L_1	.150		3.81		
Q	.015	.060	.38	1.52	5
S		.098		2.49	8
S_1	.005		.13		8
α	0°	15°	0°	15°	

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Index area: A notch or pin one identification mark shall be located adjacent to pin one and shall be located within the shaded area shown. The manufacturer's identification shall not be used as a pin one identification mark.
4. The minimum limit for dimension b_1 may be .023 inch(.58 mm) for leads number 1, 7, 8, and 14 only.
5. Dimension Q shall be measured from the seating plane to the base plane.
6. This dimension allows for off-center lid, meniscus, and glass overrun.
7. The basic pin spacing is .100 inch (2.54 mm) between centerlines. Each pin centerline shall be located within ±.010 inch (.25 mm) of its exact longitudinal position relative to pins 1 and 14 (see figure 6).
8. Applies to all four corners (leads number 1, 7, 8, and 14).
9. Lead center when α is 0°. E_1 shall be measured at the centerline of the leads.
10. All leads.
11. Twelve spaces.
12. No organic or polymeric materials shall be molded to the bottom of the package to cover the leads.

FIGURE 1. Dimensions and configuration for type 2N6989 - Continued.

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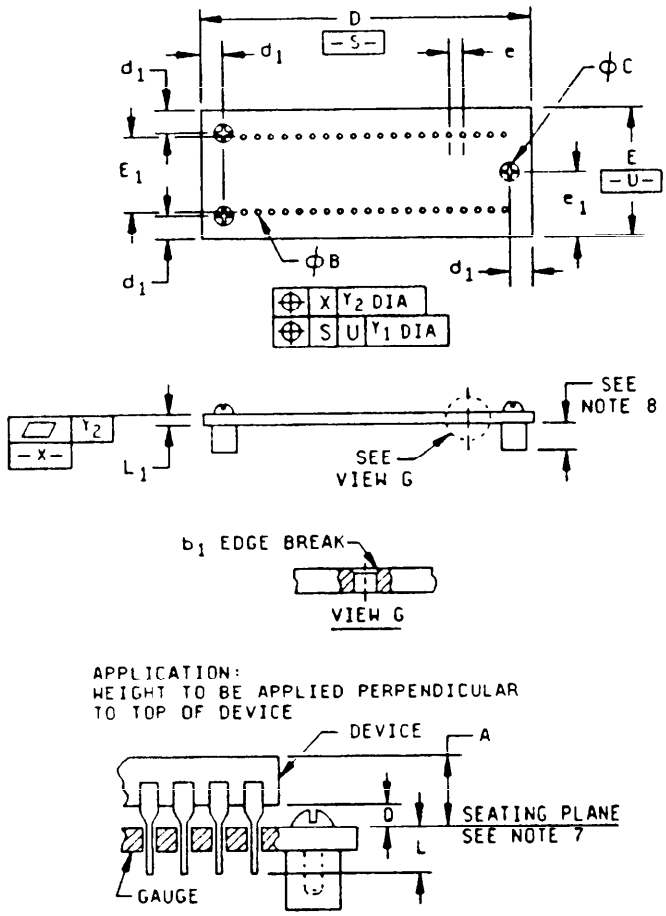


FIGURE 2. Lead gauge.

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Symbol	Dimensions				Notes
	Inches		Millimeters		
	Min	Max	Min	Max	
ϕB	.0323	.0333	.82	.85	
b_1		.002		.05	
ϕC	.119	.129	3.02	3.28	
D	1.750		44.45		5
d_1	.200 NOM		5.08 NOM		
E	.750		19.05		5
E_1	.300 BSC		7.62 BSC		
e	.100 BSC		2.54 BSC		
e_1	.375		9.53		5
L_1	.090	.099	2.28	2.51	
N	14		14		4
Y_1		.010		.25	
Y_2		.002		.05	

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Dimensioning and tolerancing in accordance with ANSI Y14.5.
4. N is the number of ϕB lead position holes in each gauge variation.
5. Maximum dimensions at D, E, and e_1 are manufacturing options with e_1 = E/2.
6. The seating plane shall be established by inserting the terminal leads of a device into the gauge with a weight of 8 mg.(min) to 12 mg.(max) (per lead) applied perpendicular to the top of the device. Dimensions A, Q, and L (see figure 1) can then be measured.
7. Height of leg optional, but sufficient to prevent lead of inserted device from bottoming on measuring table.
8. Controlling dimensions: Inch.

FIGURE 2. Lead gauge - Continued.

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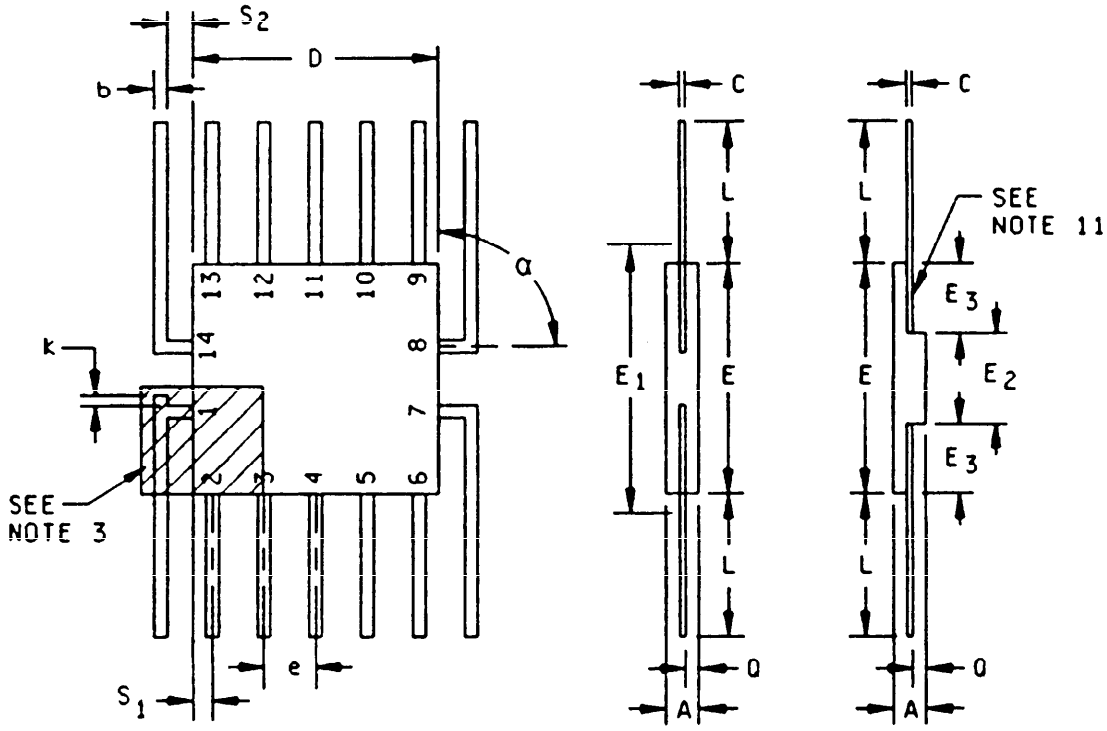


FIGURE 3. Physical dimensions for type 2N6990.

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Symbol	Dimensions				Notes
	Inches		Millimeters		
	Min	Max	Min	Max	
A	.030	.085	.76	2.18	
b	.010	.019	.25	.48	7
c	.003	.006	.08	.15	7
D		.280		7.11	5
E	.240	.260	6.10	6.60	
E_1		.290		7.37	5
E_2	.125		3.18		

Symbol	Dimensions				Notes
	Inches		Millimeters		
	Min	Max	Min	Max	
E_3	.030		.76		
e	.050 BSC		1.27 BSC		6, 8
k	.008	.015	.20	.38	12
L	.250	.370	6.35	9.40	
Q	.005	.040	.13	1.02	4
S_1	.005		.13		9,10
S_2	.004		.10		13
α	30°	90°	30°	90°	14

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Index area: A notch or pin one identification mark shall be located adjacent to pin one and shall be located within the shaded area shown. The manufacturer's identification shall not be used as a pin one identification mark. Alternatively, a tab (dim k) may be used to identify pin one.
4. Dimension Q shall be measured at the point of exit of the lead from the body.
5. This dimension allows for off-center lid, meniscus, and glass overrun.
6. The basic pin spacing is .050 inch (1.25 mm) between centerlines. Each pin centerline shall be located within ±.005 inch (.13 mm) of its exact longitudinal position relative to pins 1 and 14.
7. All leads: Increase maximum limit by .003 inch (.08 mm) measured at the center of the flat when the lead finish is solder.
8. Twelve spaces.
9. Applies to all four corners (leads number 2, 6, 9, and 13).
10. Dimension S_1 may be .000 inch (.00 mm if leads number 2, 6, 9, and 13) bend toward the cavity of the package within one lead width from the point of entry of the lead into the body or if the leads are brazed to the metallized ceramic body.
11. No organic or polymeric materials shall be molded to the bottom of the package to cover the leads.
12. Optional, see note 1. If a pin one identification mark is used in addition to this tab, the minimum limit of dimension k does not apply.
13. Applies to leads number 1, 7, 8, and 14.
14. Lead configuration is optional within dimension E except dimensions b and c apply.

FIGURE 3. Physical dimensions for type 2N6990 - Continued.

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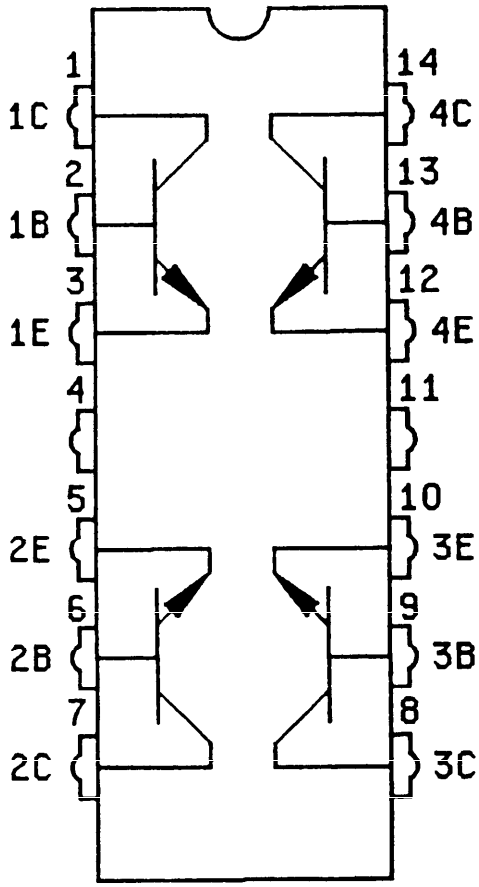
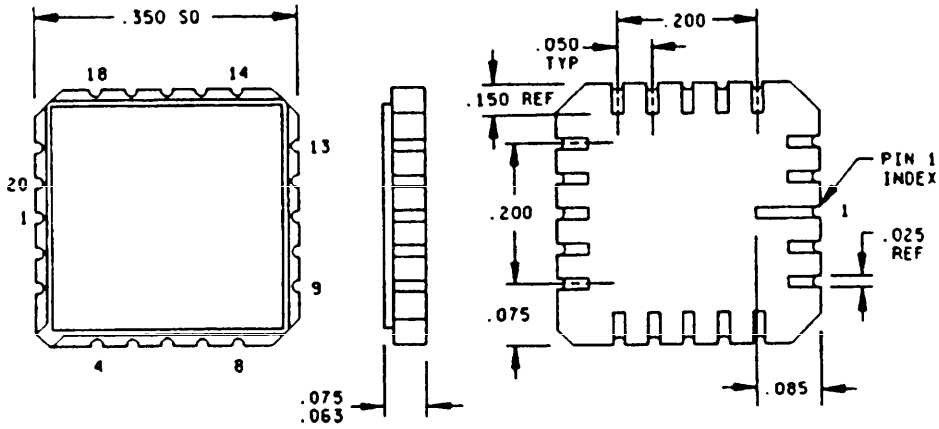


FIGURE 4. Schematic and terminal connections for type 2N6989 and 2N6990.

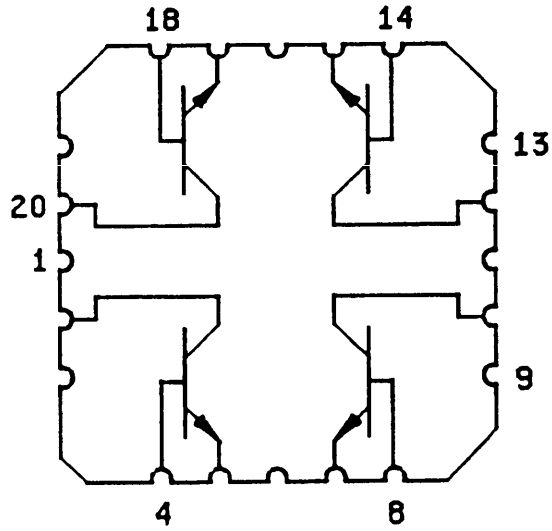
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Inches	mm
.025	0.64
.050	1.27
.063	1.60
.075	1.90
.085	2.16
.150	3.81
.200	5.08
.350	8.89

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Unless otherwise specified, tolerance is ±.005 inch (0.13 mm)

FIGURE 5. Physical dimensions for type 2N6989U.



TOP VIEW

FIGURE 6. Schematic and terminal connections for type 2N6989U.

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2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATION

MILITARY

MIL-S-19500 - Semiconductor Devices, General Specification for.

STANDARD

MILITARY

MIL-STD-750 - Test Methods for Semiconductor Devices.

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Defense Printing Service Detachment Office, Building 4D (Customer Service), 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

2.2 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Associated detail specification. The individual item requirements shall be in accordance with MIL-S-19500 and as specified herein.

3.2 Abbreviations, symbols, and definitions. Abbreviations, symbols, and definitions used herein shall be as specified in MIL-S-19500.

3.3 Design, construction, and physical dimensions. The design, construction, and physical dimensions shall be as specified in MIL-S-19500, and on figures 1, 3, and 5 herein.

3.3.1 Lead finish. Lead finish shall be solderable in accordance with MIL-S-19500.

3.3.2 Schematic and terminal connections. The schematic and terminal connections shall be as shown on figure 4 (for flat package and dual-in-line) and on figure 6 (for leadless chip carrier).

3.4 Marking. Marking shall be in accordance with MIL-S-19500.

4. QUALITY ASSURANCE PROVISIONS

4.1 Sampling and inspection. Sampling and inspection shall be in accordance with MIL-S-19500 and as specified herein.

4.2 Qualification inspection. Qualification inspection shall be in accordance with MIL-S-19500.

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4.3 Screening (JANS, JANTX, and JANTXV levels only). Screening shall be in accordance with table II of MIL-S-19500, and as specified herein. The following measurements shall be made in accordance with table I herein. Devices that exceed the limits of table I herein shall not be acceptable.

Screen (see table II of MIL-S-19500)	Measurements	
	JANS level	JANTX and JANTXV levels
7(a) or 14(a)	Maximum leak rate = 1 x 10^{-7} atm cc/s Test condition C, step 1 only	
11	I_{CBO1} and h_{FE4};	I_{CBO1} and h_{FE4}
12	See 4.3.1	See 4.3.1
13	Subgroups 2 and 3 of table I herein; ΔI_{CBO1} = 100% of initial value or 5 nA dc, whichever is greater; Δh_{FE4} = ±15% of initial value.	Subgroup 2 of table I herein; ΔI_{CBO1} = 100% of initial value or 5 nA dc, whichever is greater; Δh_{FE4} = ±15% of initial value.

4.3.1 Power burn-in conditions. Power burn-in conditions are as follows:

T_A = +25°C ± 3°C; V_{CB} = 30 V dc; P_T = 1.5 W (2N6989); P_T = 1 W (2N6989U); P_T = 0.4 W (2N6990).

NOTE: No heat sink or forced air cooling on the devices shall be permitted.

4.4 Quality conformance inspection. Quality conformance inspection shall be in accordance with MIL-S-19500.

4.4.1 Group A inspection. Group A inspection shall be conducted in accordance with MIL-S-19500 and table I herein.

4.4.2 Group B inspection. Group B inspection shall be conducted in accordance with the conditions specified for subgroup testing in IVa (JANS) and table IVb (JANTX and JANTXV) of MIL-S-19500. Electrical measurements (end-points) shall be in accordance with table I, group A, subgroup 2 herein.

4.4.2.1 Group B inspection, table IVa (JANS) of MIL-S-19500.

Subgroup	Method	Condition
B4	1036	V_{CB} = 10 V dc; P_T = 1.5 W, (2N6989), P_T = 0.4 W (2N6990), P_T = 1 W (2N6989U) at T_A = +25°C ± 3°C; t_{on} = t_{off} = 3 minutes minimum for 2,000 cycles. No heat sink or forced-air cooling on devices shall be permitted.
B5	1027	V_{CB} = 10 V dc, T_A = +125°C ±25°C for 96 hours; P_T = 1.5 W, (2N6989), P_T = 0.4 W, (2N6990) at T_A = +100°C or adjusted as required by the chosen T_A to give an average lot T_J = +275°C.
B6	3131	$R_{\theta JA}$ = 116°C/W for 2N6989 and 2N6989U, $R_{\theta JA}$ = 437°C/W for 2N6990; 22 devices, c = 0; test one transistor per device.

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4.4.2.2 Group B inspection, table IVb (JANTX and JANTXV) of MIL-S-19500.

Subgroup	Method	Condition
B3	1036	V_{CB} = 10 V dc; P_T = 1.5 W, (2N6989), P_T = 0.4 W (2N6990), P_T = 1 W (2N6989U) at T_A = +30°C ±5°C. No heat sink or forced-air cooling on the devices shall be permitted.
B5	3131	$R_{\theta JA}$ = 116°C/W for 2N6989 and 2N6989U, $R_{\theta JA}$ = 437°C/W for 2N6990; 22 devices, c = 0; test one transistor per device.

4.4.3 Group C inspection. Group C inspection shall be conducted in accordance with the conditions specified for subgroup testing in table V of MIL-S-19500. Electrical measurements (end-points) and delta requirements shall be in accordance with table I, group A, subgroup 2 herein.

Subgroup	Method	Condition
C2	2036	Test condition E, 3 oz weight; three bends of 15° for 2N6990; three bends for 2N6989; not applicable to 2N6989U.
C6	1026	V_{CB} = 10 V dc; P_T = 1.5 W (2N6989), P_T = 0.4 W (2N6990), P_T = 1 W (2N6989U) at T_A = +30°C ±5°C. No heat sink or forced-air cooling on device shall be permitted.

4.5 Method of inspection. Methods of inspection shall be as specified in the appropriate tables and as follows.

4.5.1 Pulse measurements. Conditions for pulse measurement shall be as specified in section 4 of MIL-STD-750.

4.5.2 Input capacitance. This test shall be conducted in accordance with MIL-STD-750, method 3240, except the output capacitor shall be omitted.

4.5.3 Independent transistor inspections. Inspections shall be performed on each transistor in the array.

4.5.4 Transistor-to-transistor resistance. The leads of each transistor shall be shorted together for this test. The resistance shall be measured between each transistor in the array.

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TABLE I. Group A inspection.

Inspection 1/	MIL-STD-750		Symbol	Limits		Unit
	Method	Conditions		Min	Max	
Subgroup 1						
Visual and mechanical examination	2071					
Subgroup 2						
Breakdown voltage collector to base	3001	Bias condition D; I_C = 10 µA dc	$V_{(BR)CBO}$	75		V dc
Breakdown voltage emitter to base	3026	Bias condition D; I_E = 10 µA dc	$V_{(BR)EBO}$	6		V dc
Breakdown voltage collector to emitter	3011	Bias condition D; I_C = 10 mA dc; pulsed (see 4.5.1)	$V_{(BR)CEO}$	50		V dc
Collector to base cutoff current	3036	Bias condition D; V_{CB} = 60 V dc	I_{CBO1}		10	nA dc
Emitter to base cutoff current	3061	Bias condition D; V_{EB} = 4 V dc	I_{EBO}		10	nA dc
Forward-current transfer ratio	3076	V_{CE} = 10 V dc; I_C = 0.1 mA dc	h_{FE1}	50		
Forward-current transfer ratio	3076	V_{CE} = 10 V dc; I_C = 1.0 mA dc	h_{FE2}	75	325	
Forward-current transfer ratio	3076	V_{CE} = 10 V dc; I_C = 10 mA dc	h_{FE3}	100		
Forward-current transfer ratio	3076	V_{CE} = 10 V dc; I_C = 150 mA dc; pulsed (see 4.5.1)	h_{FE4}	100	300	
Forward-current transfer ratio	3076	V_{CE} = 10 V dc; I_C = 500 mA dc; pulsed (see 4.5.1)	h_{FE5}	30		
Collector to emitter voltage (saturated)	3071	I_C = 150 mA dc; I_B = 15 mA dc pulsed (see 4.5.1)	$V_{CE(sat)1}$		0.3	V dc
Collector to emitter voltage (saturated)	3071	I_C = 500 mA dc; I_B = 50 mA dc; pulsed (see 4.5.1)	$V_{CE(sat)2}$		1.0	V dc
Base emitter saturated voltage	3066	Test condition A; I_C = 150 mA dc; I_B = 15 mA dc; pulsed (see 4.5.1)	$V_{BE(sat)1}$	0.6	1.2	V dc

See footnote at end of table.

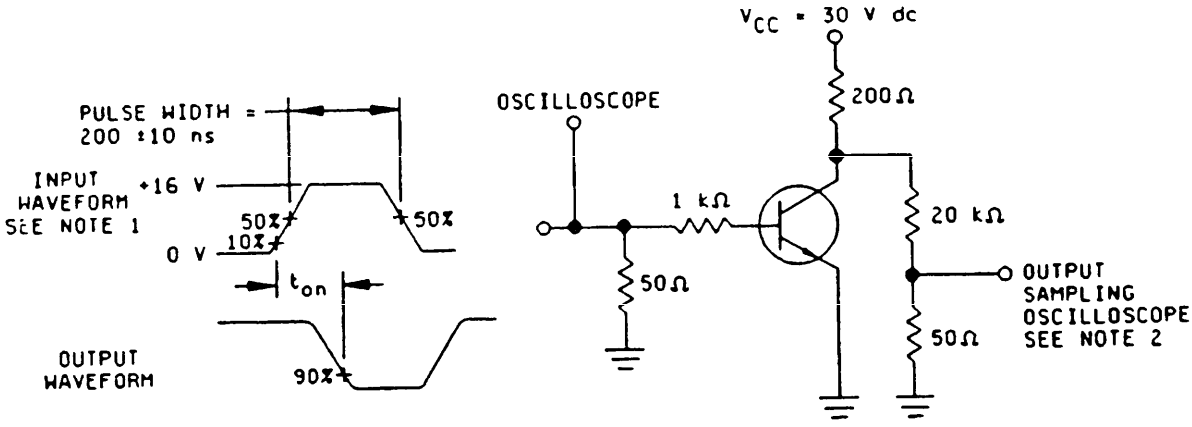
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TABLE I. Group A inspection - Continued.

Inspection 1/	MIL-STD-750		Symbol	Limits		Unit
	Method	Conditions		Min	Max	
Subgroup 2 - Continued						
Base to emitter saturated voltage	3066	Test condition A; I_C = 500 mA dc; I_B = 50 mA dc; pulsed (see 4.5.1)	$V_{BE(sat)2}$		2.0	V dc
Subgroup 3						
High temperature operation:		T_A = +150°C				
Collector to base cutoff current	3036	Bias condition D; V_{CB} = 60 V dc	I_{CBO2}		10	µA dc
Low temperature operation:		T_A = +125°C				
Forward-current transfer ratio	3076	V_{CE} = 10 V dc; I_C = 10 mA dc	h_{FE6}	35		
Subgroup 4						
Small-signal short-circuit forward-current transfer ratio	3206	V_{CE} = 10 V dc; I_C = 1 mA dc; f = 1 kHz	h_{fe}	50		
Magnitude of small-signal short-circuit forward-current transfer ratio	3306	V_{CE} = 10 V dc; I_C = 20 mA dc; f = 100 MHz	$\|h_{fe}\|$	2.5	8.0	
Open circuit output capacitance	3236	V_{CB} = 10 V dc; I_E = 0; 100 kHz ≤ f ≤ 1 MHz	C_{obo}		8.0	pF
Input capacitance (output open-circuited)	3240	V_{EB} = 0.5 V dc; I_E = 0; 100 kHz ≤ f ≤ 1 MHz; See 4.5.2	C_{ibo}		25	pF
Turn-on time		See figure 7	t_{on}		35	ns
Turn-off time		See figure 8	t_{off}		300	ns
Transistor-to-transistor resistance		$\|V_{T-T}\|$ = 500 V dc; See 4.5.3	R_{T-T}	10^{10}		Ω

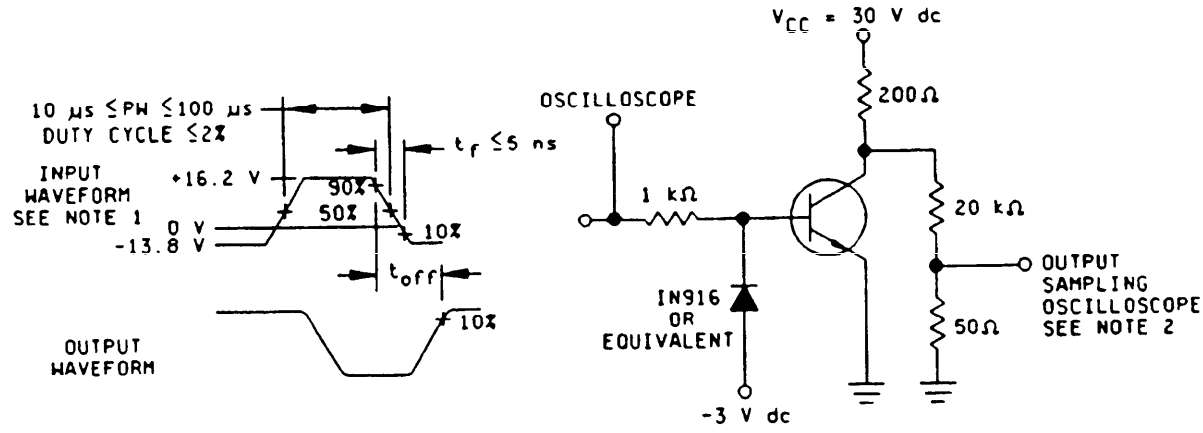
1/ For sampling plan see MIL-S-19500.

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NOTES:
1. The rise time (t_r) and fall time (t_f) of the applied pulse shall be each ≤ .2 ns; duty cycle ≤ 2 percent; generator source impedance shall be 50 Ω.
2. Output sampling oscilloscope: Z_{in} ≥ 100 kΩ; C_{in} ≤ 12 pF; rise time ≤ 5.0 ns.

FIGURE 7. Saturated turn-on switching time test circuit.



NOTES:
1. The rise time (t_r) and fall time (t_f) of the applied pulse shall be each ≤ .2 ns; duty cycle ≤ 2 percent; generator source impedance shall be 50 Ω.
2. Output sampling oscilloscope: Z_{in} ≥ 100 kΩ; C_{in} ≤ 12 pF; rise time ≤ 5.0 ns.

FIGURE 8. Saturated turn-off switching time test circuit.

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5.1 Packaging requirements. The requirements for packaging shall be in accordance with MIL-S-19500.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Notes. The notes specified in MIL-S-19500 are applicable to this specification.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Issue of DODISS to be cited in the solicitation.

b. Lead finish (see 3.3.1).

c. Type designation and product assurance level.

6.3 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

CONCLUDING MATERIAL

Custodians:
Army - ER
Navy - EC
Air Force - 17

Review activities:
Army - AR, MI, SM
Navy - AS, CG, MC
Air Force - 13, 15, 19, 85, 99

Preparing activity:
DLA - ES

(Project 5961-1505)

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.

2. The submitter of this form must complete blocks 4, 5, 6, and 7.

3. The preparing activity must provide a reply within 30 days from receipt of the form.

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I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER MIL-S-19500/559B	2. DOCUMENT DATE (YYMMDD) 94/04/29

3. DOCUMENT TITLE SEMICONDUCTOR DEVICE, UNITIZED, NPN, SILICON, SWITCHING, FOUR TRANSISTOR ARRAY TYPES 2N6989, 2N6989U, AND 2N6990, JANTX, JANTXV, AND JANS

4. NATURE OF CHANGE (Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (Last, First, Middle initial)	b. ORGANIZATION	
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON (If applicable)	7. DATE SUBMITTED (YYMMDD)

8. PREPARING ACTIVITY

a. NAME Alan Barone	b. TELEPHONE (Include Area Code) (1) Commercial 513-296-6048	(2) AUTOVON 986-6048
c. ADDRESS (Include Zip Code) Defense Electronics Supply Center ATTN: DESC-ELDT Dayton, OH 45444-5765	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340	

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